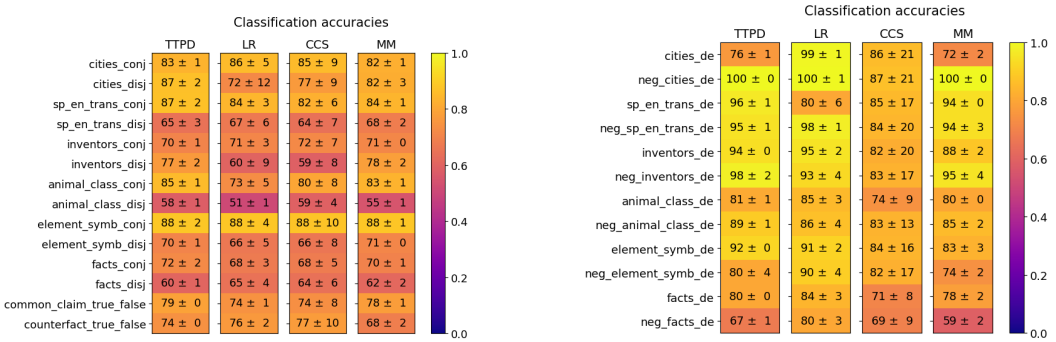


(a) (b)

Figure 9: Generalization accuracies of TTPD, LR, CCS and MM. Mean and standard deviation are computed from 20 training runs, each on a different random sample of the training data.

In the second failure mode, the learned truth direction was not able to accurately separate true vs. false statements. This failure mode occurred in *inventors*, *neg_inventors* and probably also in *facts* and *neg_facts*. Example statements include:

- Ernesto Blanco did not live in the U.S. (False)
- Gideon Sundback did not live in the U.S. (True)
- The atomic number of an element represents the number of electrons in its nucleus. (False)

In the real-world scenarios, the main failure mode seems to be the bias that fails to generalize. Lies and truthful replies separate perfectly along $\mathbf{t}_G$ with an AUROC of $\approx 1.00$. However, the classification accuracy of TTPD is not 100%, and out of $\sim$8 misclassified statements, 6-8 are lies. This suggests a generalisation failure of the bias.

The Logistic Regression classifier also has these two failure modes (bias fails to generalize, truth direction fails to generalize), but compared to TTPD it is less often the bias that fails to generalise and more often the truth direction. The lies and truthful responses from the real-world scenarios separate along $\mathbf{d}_{LR}$, the direction of the LR classifier, with an AUROC of only $\approx 0.86$ and out of $\sim$22 misclassified real-world scenarios, $\sim$16 are false and $\sim$6 are true. This suggests that mainly the truth direction $\mathbf{d}_{LR}$ fails to generalize. We hypothesise that this difference between TTPD and LR arises because LR learns bias and truth direction at the same time, whereas TTPD learns the truth direction first and then the bias. In summary, it seems that a truth direction that is learned separately from the bias generalises better, at the cost that it is harder to find a well-generalizing bias.

# E Generalization to logical conjunctions, disjunctions and statements in German

This section provides a detailed breakdown of the classification accuracies for TTPD, LR, CCS, and MM on individual datasets comprising logical conjunctions, disjunctions, and German statements. Figure 9 presents these results in full, complementing the summarised view shown in Figure 6b of the main text. It is important to note that all classifiers were trained exclusively on activations from English affirmative and negated statements.

# F More details on the real-world scenarios

## F.1 Generating the LLM responses to the scenarios

The responses are generated by iteratively sampling the next token using the softmax probabilities derived from the model's logits, corresponding to a temperature setting of $T = 1$. We interrupt the generation after the first period (".") or exclamation mark ("!") since these typically mark the end of a sentence.